

Jay Niketan Pathare

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EXPERIENCE

Graduate Student Developer | HeinOnline (CSE 611 Project), Buffalo, NY

January 2026 – April 2026

- Engineered a fault-tolerant OCR/LLM metadata extraction pipeline (Tesseract-OCR, Qwen-VL:8b, Qwen-35B) across three independent sources: CrossRef API, web scrapers, and OCR/LLM vision to extract article titles, authors, DOIs, page ranges, and ToC data from law journals (30k+ pages).
- Built a fuzzy-matching engine to merge outputs into a validated structure, cutting manual data-entry work by ~90% and pushing extraction accuracy to ~91% over five iterations, benchmarked against ground truth.
- Parallelized the pipeline with a worker pool that pulls the next journal from a shared queue on completion, and added an “aiscore” system to flag low-confidence fields for human review.

Software Engineer | Thesis Mumbai Tech (Startup), Mumbai, India

August 2024 – August 2025

- Led delivery of platform modules end to end: gathered requirements in client meetings, prototyped to validate scope, and broke the finalized work down across a small team.
- Reviewed and merged team pull requests alongside my own code, catching issues and maintaining code quality.
- Built and scaled healthcare modules (Patient Management, Consent) supporting 10k+ records on normalized relational schemas for fast retrieval and reporting, and designed a real-time WebSocket + Redis pipeline feeding IoT baby-warmer sensor data into PostgreSQL with ~2s live monitoring.
- Delivered a real-time telemedicine consultation feature (WebRTC) connecting on-site nurses with remote doctors to review test results and procedures with patients, alongside secure PDF report generation, and added a color-blindness diagnostic module (95%+ accuracy) with automated reporting for practitioners.
- Containerized 3+ production projects using Docker and Docker Compose, keeping local, staging, and deployment environments consistent and reducing new-developer setup time by ~90%.

Cloud Engineer Intern | Data Maven Pvt Ltd, Mumbai, India

November 2023 – May 2024

- Designed and deployed scalable AWS infrastructure (EC2, RDS) with VPC networking and security groups for secure, low-latency communication between backend services.
- Automated cloud resource provisioning, cutting setup time by ~40% and reducing deployment issues.

Data Engineer Intern | Go Digital Technology Consulting, Mumbai, India

June 2023 – August 2023

- Developed an ETL workflow using Python (Pandas, NumPy) and MySQL to process and clean large business datasets, producing reports on key operational trends.
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ACADEMIC PROJECTS

LedgerFlow: Event-Driven Payment Ledger | Java, Spring Boot, Kafka, PostgreSQL

October 2026

- Implemented a payment ledger that posts each payment exactly once even when messages are duplicated or retried, using a transactional outbox, idempotent consumers, and double-entry accounting.
- Prevented overspending under concurrent requests with ordered PostgreSQL row locks: in a 200-request race on an account funded for 50 payments, exactly 50 succeeded, 150 were safely rejected, and the balance ended at \$0.
- Verified with 94 automated tests and a 27/27-run chaos campaign (18,000 duplicate events, 14 real JVM crashes, 0 invariant violations); a local Kubernetes (k3d) pod-restart demo completed 250/250 payments with 0 duplicate postings.

RIPPLE: Agentic Change-Impact Analysis | Python, Pydantic, Git, LLM API

September 2026

- Developed RIPPLE, an AI agent (Python CLI) that predicts which files, tests, and regression risks a feature request will touch before any code is written, using 7 static-analysis tools, a 25-tool-call cap, and Python-verified evidence.
- Evaluated on 30 unseen software-change tasks from 16 repositories (90 runs), reaching 0.306 precision and 0.311 MRR, with 0 unsupported claims accepted into any report.

LLM Application Optimization Engine | Python, Qwen (Novita API), Groq API, pytest, LaTeX

August 2026

- Designed a multi-model pipeline that turns a job description into a tailored one-page resume, with Python checking every generated claim against source evidence and rejecting unsupported ones, backed by a 2,500+ test pytest suite.
 - Cut typical end-to-end runtime by ~73% (about 11 minutes to about 3) using three concurrent generation workers, regeneration of only failed parts, and local fixes in place of repeat model calls.
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EDUCATION

Master's in Computer Science

August 2025 – February 2027

University at Buffalo, The State University of New York, Buffalo, NY

GPA: 3.85 / 4.0

Bachelor of Engineering, Information Technology

Vivekanand Education Society's Institute of Technology, Mumbai University, India

August 2019 – May 2023

TECHNICAL SKILLS

- Languages & Databases:** Python, Java, JavaScript, Go, SQL, PostgreSQL, MySQL
 - Frameworks:** Django, Django REST Framework, Spring Boot, ReactJS, Pandas, NumPy, PyTorch, scikit-learn, Pydantic
 - AI/ML:** LLM Integration, Prompt Engineering, Machine Learning, Deep Learning, Agentic AI, AI Agents, Tool Calling, Model Evaluation, Neural Networks (CNN, GNN), Transformers
 - Software Engineering:** OOP, REST APIs, Full-Stack Development, Backend Development, System Design, ETL, Parallel Processing, Agile, Real-Time Monitoring, Event-Driven Architecture, Distributed Systems, Apache Kafka
 - Tools:** Linux, Redis, AWS (EC2/RDS), Docker, Git, GitHub, GitHub Copilot, WebSockets, OAuth/Auth0, Kubernetes
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CERTIFICATIONS, AWARDS & PUBLICATIONS

- 1st Place, ACV Auctions + Copart Challenge** | AI for Good Hackathon Design Challenge, UB October 2026
- AWS Certified Cloud Practitioner** July 2024
- Research Publication:** J. Pathare et al., “Music Genre Classification,” IJRAR, Apr. 2023 (97.68% accuracy). [Link](#).